IGI LIMS

Feature overview of the specimen tracking system, as built to date

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1. Purpose

IGI LIMS is a chain-of-custody database and web interface for the biological specimens collected in the IGI cancer clinical trial. It records one row per tube. Every tube stores the tube it was made from, so the path from a collected specimen, through fractionation, cryopreservation, extraction and library preparation, to the sequencing or mass-spectrometry data file is a chain of parent links that can be walked in either direction. The system runs as a single Flask application on SQLite with no external services, suitable for a small lab (about two dozen users) on a workstation or a GCP virtual machine.

2. Tube identifiers

Identifiers are composed by the database from the tube's attributes. They are never typed by hand. The format is defined in one place (`config.py`, `ID_SEGMENTS`) so that it can change as the specification settles without touching the schema.

```
IGI-CC-PPPPPPP-YYMMDD-SS-FFF-MMM-AA (35 characters)
```

Segment	Meaning	Validation
IGI	Project constant	literal
CC	Cohort	active code in the cohort list (MP, AM, TN, NC)
PPPPPPP	Participant code	seven upper-case letters, no I or O
YYMMDD	Collection date	parsed as a calendar date
SS	Specimen type	active code in the specimen list (PF, PB, BM, TU, LN, ...)
FFF	Fraction	active code in the fraction list (BLK, PLA, PBM, CD8, ...)
MMM	Material	active code in the material list (NAT, CPU, DNA, LIB, PEP, ...)
AA	Aliquot number	01 to 99, issued per prefix by the database, never reused

The aliquot counter is taken over all tubes including deleted ones, so an ID can never refer to two different physical tubes. The anchored regular expression for the current format is published at `/api/pattern` for pipelines and validation scripts.

3. What the system enforces

Controlled vocabularies

Every code used in IDs, forms and metadata lives in one code-list table and is managed on the Codes page. Codes are retired rather than deleted, so historical records keep their meaning. Currently 46 categories are defined: the core ID vocabularies (cohort, specimen, fraction, material), status, assay, biobank source, site, unit, and admin-extendable lists for SOPs, freezers, racks, boxes, positions, custody parties and purposes, kits, lots, buffers, media, antibodies, instruments and so on. Administrators can add new categories without a code change.

No free text except Notes

Every tube, custody and metadata field other than the Notes box is a dropdown backed by one of these lists. The operator must be an active user account. Server-side checks reject any value that is not an active entry, so the database cannot drift into inconsistent spellings. The only remaining free-text fields are machine identifiers copied from pipeline output (data file URI, run ID, checksum) and the participant's external site ID.

Derivation rules

A rule table (Rules page, 140 rules seeded as a starting proposal) decides which fractions a specimen type may have, which fraction-to-fraction and material-to-material transitions are allowed, which materials a root tube may arrive as, and which fractions a material may carry. A violation blocks creation and offers a Request override button instead of silently accepting the record.

Two-person agreement

Overrides of derivation rules, and by default every edit, delete or restore of a sample record, go to an Approvals queue. The requester cannot approve their own request. The approver's decision applies the change, and the audit log records the approval ID alongside the change. The scope can be relaxed in configuration to provenance-only changes (parent, ID components, delete) or switched off.

Soft deletes and full audit

Deleting a tube keeps the record, keeps the aliquot number reserved, and is refused while the tube has active children. Every write (participants, samples, custody, files, metadata, attachments, code lists, users) is recorded in the audit log with who, when, and a before and after snapshot shown as key and value lines.

4. Registration workflow

The Register landing page offers exactly three actions and nothing else, to keep one obvious path for each task.

- **Participant.** Create a participant with cohort, generated participant code, and external site ID.
- **Base specimen.** Register a collected tube (a root with no parent): specimen type, collection date, fraction and material on arrival, biobank source (CCPD, HemaTrack, in-house, or not applicable), operator, SOP, quantity and unit, storage location, status, and notes. All fields are required; lists carry an NA entry where the question does not apply.
- **Derived specimen.** Choose the participant, then the fraction and material of the tube being made. The system lists only the parent tubes from that participant that the derivation rules accept as inputs, and shows the ineligible ones with the reason. Creating the child can mark the parent as depleted in the same step.

Each tube page shows its components, parent and children, custody events, data files, attachments, metadata, and history. Editing goes through the approval queue.

5. Browsing: Samples & Participants

A single read-only page. Pick a participant and the page draws a tall, top-down tree of every collection and everything derived from it. Hovering a node shows its status, quantity, location and key metadata; clicking pins that panel with a link to the full tube page. Nothing on this page creates or changes records, so it is safe to

leave open on a shared screen.

6. Metadata per fraction and material

Metadata fields are defined in one table (Fields page) and attached to a fraction or a material, so a tube shows only the fields that make sense for what it is. The create and edit forms render these fields dynamically, the tube page and the tree hover panel display them, and they are included in the audit diff. Fields are typed as number (with unit), fixed choice, admin-extendable list, yes/no, date, or date-time. There is no text type. Two fields are required at present: the assay of a sequencing library and the HLA class of a peptide eluate. 199 field definitions are active today, for example:

Scope	Examples
Plasma, serum, supernatant	spin protocol, hemolysis grade, time to processing, filter
PBMC, PFMC, buffy coat	isolation method, viable cell count, viability, hashtag, cryo medium, freeze method
Immune, tumor, T-cell fractions	enrichment or selection method, marker panel, purity, input cells, kit and lot
DNA, RNA, HMW DNA/RNA, cfDNA	extraction kit, concentration, A260/280, A260/230, RIN or DIN, fragment size, elution
Sequencing library	assay, library prep kit, capture kit, index, shearing, molarity, submission
Peptide eluate	HLA class, antibody and lot, bead type, input cells, elution, cleanup, instrument
Tissue (snap-frozen, OCT, FFPE)	block ID, dissociation method, input tissue mass

7. QC file import and attachments

Any file (trace, image, report) can be attached to a tube; it is stored on disk under the attachment directory with an md5 checksum. The Import QC page goes further for common instrument exports: upload one file, the system parses it, shows one row per instrument sample with a suggested tube ID (matched automatically when the sample name contains a tube ID), and writes the ticked values into the tube's metadata fields. Each imported value keeps a link back to the stored export file, so the provenance of a concentration or RIN can be checked later. Imports are direct and audited; manual metadata edits still go through approval.

Instrument	Format	Values captured
Qubit 4	CSV, parsed through allotropy	concentration (original and diluted), assay, units
NanoDrop One	CSV, parsed through allotropy	concentration, A260/280, A260/230
TapeStation	XML	concentration, RIN or DIN, region average size, molarity, percent of total
Countess 3	CSV, columns from the vendor guide	total, live and dead concentration, viability, cells counted, average size, aggregation

8. Custody and data files

Custody events are append-only: from and to (a user or an external party such as HTSF or CCPD), purpose (extraction, sequencing submission, storage, immunopeptidomics, culture, sorting, return, shipment), timestamp, and condition on receipt. Data files, which live in Google Cloud Storage buckets, are linked to the tube that was sequenced or run (usually a library or peptide eluate) by URI, with assay, run ID, checksum and pipeline version. A file is therefore always reachable from the collected specimen it came from.

9. Users, roles and administration

- **Roles.** admin (everything), user (create, derive, edit, approve), viewer (read only). Passwords are hashed; accounts are created and reset from the command line or the Users page.
- **Codes.** Add, rename or retire entries in any list; add a whole new list; see which fields use each list.
- **Rules.** View and edit the derivation rule table.
- **Fields.** Add metadata fields of any supported type, mark them required, or retire them.
- **Audit.** Searchable log of every change.

10. Read-only JSON API

For pipelines and scripts. Authenticate with a browser session or an `X-Api-Token` header matching the token set in the server environment.

Endpoint	Returns
/api/pattern	anchored regex and fixed length for IDs
/api/validate/<id>	the ID parsed into components, and whether it exists
/api/samples?participant=&root=&material=&status=&prefix=	list of tubes
/api/samples/<id>	one tube with custody events and files
/api/samples/<id>/lineage	ancestors (root first) and all descendants
/api/tree/<root_id>	nested tree for one collection
/api/files	data files with their tube and lineage

11. Technical summary

- Python 3.10, Flask 3, Jinja templates, a small amount of vanilla JavaScript for the tree and dynamic forms.
- SQLite in WAL mode with foreign keys on; one file database, backed up by copying the file.
- Schema: code_list, derivation_rule, user, participant, sample, custody_event, data_file, approval_request, audit_log, attribute_definition, sample_attribute, attachment.
- Instrument parsing via allotropy (Benchling open source) for Qubit and NanoDrop; TapeStation and Countess parsed directly.
- 38 automated tests cover ID composition, the registration and approval workflow, parsers, import, metadata and vocabulary enforcement.
- Deployment target: a GCP VM behind gunicorn, reachable by collaborators over the web; not yet set up.

12. Open questions (tracked in NOTES.md)

- Participant code format: seven letters as specified, or a site-style code such as P101.
- NTC and NRM are three-character specimen codes but the ID segment is two characters.
- Whether every edit should require two-person approval, or only provenance changes.
- The seeded derivation rules are a proposal and need review by the lab.
- Which metadata fields should be required.

- Most admin-extendable lists start with only NA and need real freezers, racks, boxes, SOPs and kits before first use.
- The Countess parser follows the vendor column specification and has not yet been checked against a real export.